Chapter 9

Gold restorations: the metals, the manufacture and the fit

Graham Chadwick, John Radford, David Ricketts

CHAPTER CONTENTS

Gold alloys 117
Applications and constituents 117
Biocompatibility 117
Fabrication of gold restorations 117
Checking the restoration 121
Refining the casting before the patient's appointment 121
Refining the casting at the chairside 122
Cementation of the restoration 124
Luting the restoration 125
Supportive phase 127
Summary 127

GOLD ALLOYS

Applications and constituents

The restoration of teeth with gold in the form of crowns, inlays, onlays and palatal veneers has considerable historical pedigree and should still be considered in treatment planning for reasons outlined in the preceding chapter. Although, in the past, pure gold has been used to restore cavities directly by the process of cold welding, this is no longer taught as the gold foil used is considered too soft for general application.

The International Organization for Standardization (ISO) defines four types of gold alloy: 1, 2, 3 and 4. They all contain gold, silver and copper in various amounts, with platinum and palladium also present in types 2, 3 and 4. The inclusion of all these metals enhances the physical properties of the gold, rendering the alloys suitable for a wider range of clinical applications. With increasing gold alloy type number, the content of gold (85% to 65%) and silver (11% to 9%) decreases and the proportions of copper (3% to 15%) and those of platinum or palladium (2% to 10%) increase. The effects on the physical properties of the alloy, with increasing alloy type number, are to increase strength and hardness, and to decrease ductility and corrosion resistance.

Type 1 gold alloys are recommended for use in clinical situations of low functional stress such as simple inlays. Situations where an inlay involves cuspal coverage require higher stress-bearing capability and so Type 2 gold alloy is recommended. Types 3 and 4 gold alloys are used where high strength is required such as in crown and bridgework. Both of these alloy types, by virtue of their relatively high copper content, may also be used to fabricate onlays or palatal veneers where adhesion to tooth substance, by use of a chemically adhesive luting resin, is required. Such adhesion can be enhanced by, following try-in, returning the restorations to the furnace for oxidation for a period of 10 minutes at 400°C.

Biocompatibility

The biocompatibility of gold alloys is considered to be good. Although gold allergy is rare, technicians with a known sensitivity to nickel are thought to be at heightened risk of palladium allergy.

FABRICATION OF GOLD RESTORATIONS

The majority of gold restorations are manufactured indirectly in the dental laboratory. This involves a series of stages which, in industrial terms, form a total process

© 2011 Elsevier Ltd.

chain. Associated with each stage are losses in dimensional accuracy. It is important to realize that such losses may occur as both increases and decreases in dimensions; however, when the whole process chain is taken into account these balance out, culminating in a restoration that fits. A point often misunderstood by the public is that this manufacturing process is bespoke. Each restoration is made on an individual basis and is therefore unique. It is well established in general industrial production that this is the most expensive method of manufacture. The laboratory stages described for the production of a gold restoration in this chapter are common to many other materials and restorations.

Once tooth preparation has been completed by the dentist (Figure 9.1), an impression of the tooth is recorded, decontaminated and disinfected (see Chapter 13). Upon receipt of this, the dental laboratory technician casts an accurate die stone model of the preparation (Figure 9.2). Die stone is a gypsum-based material (calcium sulphate dihydrate) which has undergone a treatment process to make it harder and have a lower expansion on setting than conventional plaster. Die stones expand by 0.05–0.10% on setting compared to plaster's 0.2–0.3%. This small amount of expansion therefore produces a microscopically larger crown than the tooth preparation, ensuring a full seat at try-in.

Once the die stone has set, the technician has to section the model so that the master die of the prepared tooth can be removed to allow a wax pattern to be built up (Figures 9.2 and 9.3). It is therefore critical that the clinician clears the contact point during tooth preparation, not only to allow an impression to be taken of the margin but also to allow the model to be sectioned. Failure to do this will lead to the laboratory not being able to proceed with the work or, worse still, attempting to proceed and working on a die and contact damaged in the sectioning process. Once the model has been sectioned, the master die is covered with die spacer, a paint that is applied in coats to the master die (Figure 9.3). Two to three coats of conventional die spacer amount to a thickness of about



Figure 9.1 Occlusal (top) and buccal (bottom) view of a full gold crown preparation; the distal margin is deep. This has resulted from the operator's attempt to avoid damage to the adjacent gold restoration.

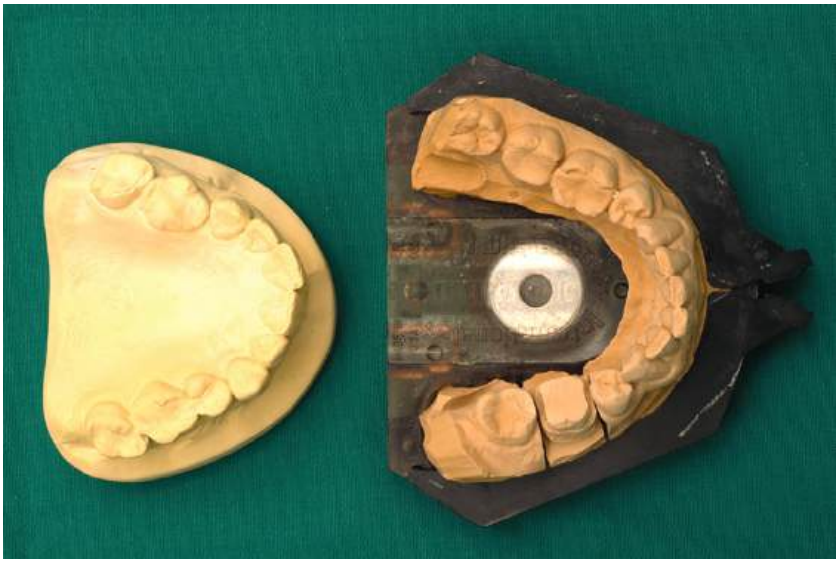


Figure 9.2 Lower die stone model cast from an impression taken of the tooth preparation seen in Figure 9.1 (right). Note the model has to be sectioned to allow the master die of the lower right first molar crown preparation to be removed from the model to build up the wax pattern. The upper model (left) is used to check the occlusion and create the occlusal form of the crown.



Figure 9.3 Sectioned model with the master die coated in die spacer.

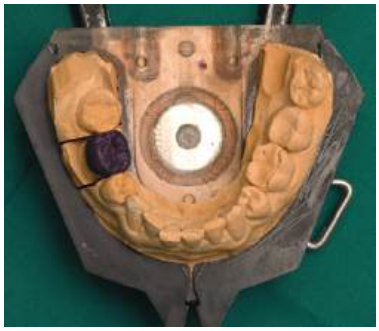


Figure 9.4 Wax pattern of a full gold crown built up on the model of the lower right first molar tooth.

25–30 μm. Ideally, the die spacer should be painted to just short of the preparation margin. The die spacer is placed to allow for an adequate layer of luting cement and full seat of the crown at fit.

Once the master die has been prepared, a wax pattern of the restoration is built up using the wax additive technique (Figure 9.4), checking that all contours are correct, all contact points are tight, and the occlusal surface and contacts conform to the existing occlusion; articulated study models facilitate this (Figure 9.5). It is also possible with the advent of computer-aided design and computer-aided manufacture (CAD-CAM; see Chapter 11) technology to computer design and mill wax blocks to form wax patterns of crowns. Once the wax pattern has been made, it is then removed from the die and a wax sprue attached (Figure 9.6). The sprued pattern is then placed

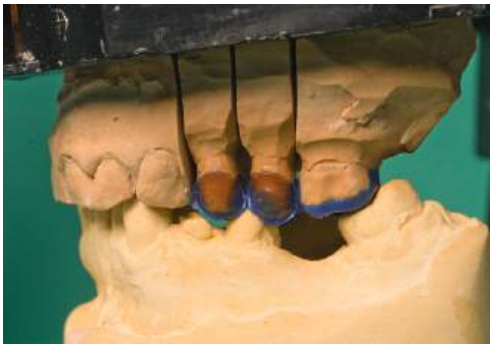


Figure 9.5 Articulated die stone model being used in the wax-up of the metal substructure for metal–ceramic crowns on the upper premolar teeth and a gold cuspal overlay on the molar tooth.



Figure 9.6 Cuspal overlay wax pattern seen in Figure 9.5 being sprued (left) and being invested in investment material (right). The sprued wax pattern is placed in an investment cylinder and the investment material is poured over – initially a small paintbrush is used (inset) to coat the wax pattern and to ensure no air bubbles attach to the wax surface.

in a lined investment cylinder and invested in an investment material which is mixed under vacuum so as to avoid the incorporation of air and hence porosities in the investment mould (Figure 9.6).

Thereafter, once the investment has set, the wax is removed from the mould by burning it off. For gold alloys this is achieved by placing the investment mould in a furnace at either 450°C (slow burn out) or 700°C (fast burn out). The mould is then placed into a casting machine and gold alloy, in molten state, is forced into it using centrifugal force. Once cooled, the surrounding investment is removed, the sprue is cut off the restoration and the resultant alloy casting trimmed and polished (Figure 9.7).

Clearly in such an elaborate process faults in the casting may occur. Common faults include:

- *Incomplete casting* – where there has been incorrect spruing or lack of either molten alloy or applied centrifugal force. This prevents the flow of sufficient gold into the investment mould.
- *Porosity and pitting* – these are seen as bubbles or pitting upon the casting surface. They may arise from porosity in the investment mould itself, the incorporation of investment material in the casting or the liberation of gases from the investment material.
- *Incorrect dimensions of restoration* – these may arise from insufficient expansion of the investment mould or other errors in the process chain.

Fitting the completed restoration to the die minimizes the chances of such a flawed casting reaching the patient for try-in.

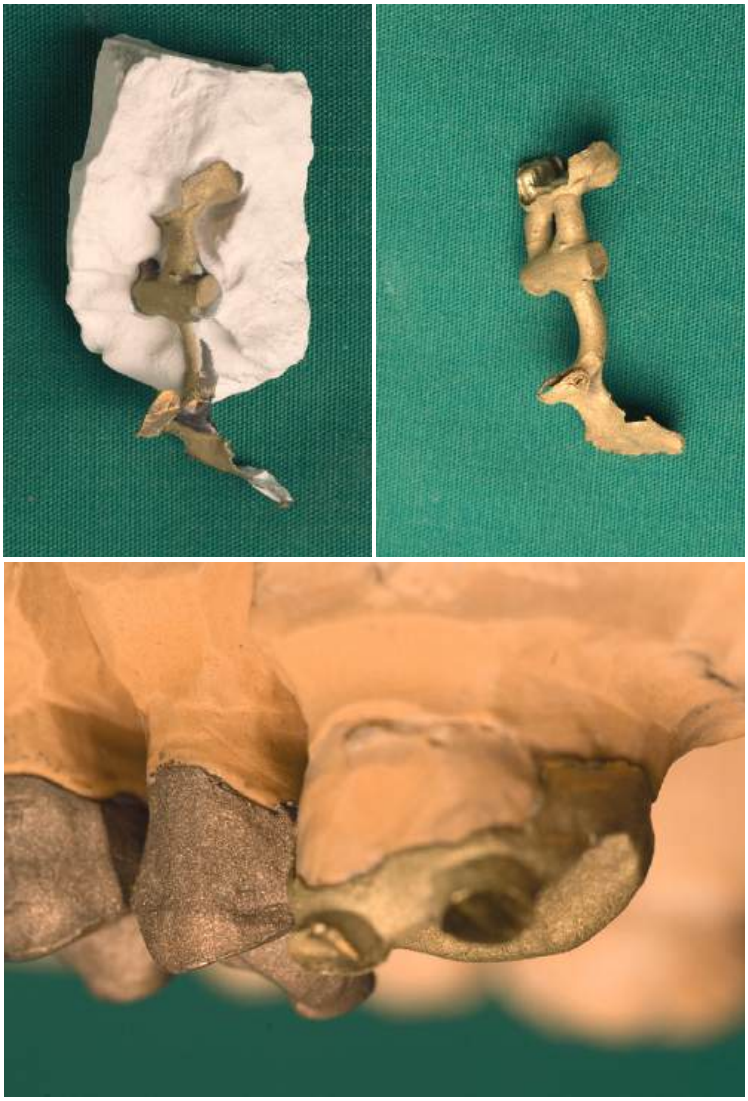


Figure 9.7 Gold casting being removed from the investment material (top left), completely removed from the investment (top right) and being tried onto the model once the sprue has been cut off (bottom).

CHECKING THE RESTORATION

On receipt of the casting, the dentist should examine it meticulously, both on and off the master die, checking for the aforementioned flaws. As the laboratory technician would have had to trim the master die in order to capture the margins, errors can occur in over- or under-trimming, leading to negative and positive ledges respectively when tried in the mouth. In addition, damage can occur to the master die during the fabrication of the restoration. Sometimes the casting may fit on the master die and not on the tooth, but this is the exception, not the rule. Similarly, damage can occur to the opposing cast resulting in occlusal discrepancies. Such errors may not be evident on the cast. Trying-in the restoration at the chairside is therefore essential before the restoration is cemented.

Refining the casting before the patient's appointment (Figure 9.8)

Prior to the patient's appointment the laboratory work should be checked for obvious errors:

- *Fit surface*. Castings with significant deficiencies or defects on the fit surface should be remade.
- *Contact areas*. The master die with the casting should be held up to the window. If daylight is visible between it and the adjacent tooth, the casting should be returned to the laboratory for a remake or possibly soldering.
- *Emergence angle*. Bulbous crowns may compromise effective home care. In order to shape these to make them more favourable, remove the casting from the working die and hold it such that the fit surface is being examined and thin the bulbosity with a green stone in a slow handpiece. An over-contoured crown is not necessarily a laboratory fault as it might occur following insufficient tooth preparation by the clinician.
- *Occlusal table*. After removing the casting from the master die, confirm centric stops on other teeth by pulling through thin articulating paper or Shimstock foil (Roeko, Germany; see later). Where there are centric stops the paper or foil will not pull through. Then place the casting on the master die and repeat the process. If there are any prematurities in intercuspal position on the restoration, the articulating paper or Shimstock foil will pull through where the previous centric stops were identified. The occlusal prematurities on the casting can then be identified using articulating paper and removed by using a stone. This is easier to do if the models have been articulated.



Figure 9.8 Fit surface of castings being checked for any defects and retained investment material. Both are satisfactory.

Refining the casting at the chairside

The most satisfactory and safest method is to try the restoration in the mouth, identify any problems or adjustments required and then remove the restoration from the mouth to carry out the adjustment.

- *Checking the marginal fit*. This should initially be assessed visually and with the use of a dental probe. The margin of the crown should seat fully with no space between the crown margin and tooth preparation margin. If a gap exists it should be established if it is localized or generalized around the whole perimeter of the restoration. In the former this is likely to be due to a local error, which could have occurred at any stage in the process from taking the impression; if this is the case, a remake is usually indicated and a new impression should be taken. The latter would indicate that the casting is not seating fully and in these circumstances the tooth preparation should be checked for any remaining temporary cement. The casting should be rechecked for bubbles of gold (where air bubbles in the investment sat at the wax pattern surface) or retained investment material. The master die should be checked for any damage, typically at any internal line angles, comparing the model with the tooth preparation; rounding internal line angles at the end of tooth preparation reduces the risk of this. Overbuilt contacts might also be the cause of the gold casting not seating fully. Tight contact points can be checked with indicator die spray such as Occlude (Pascal Co., Inc.) (Figures 9.9–9.11). This spray can be used in a similar manner to check the fit surface of the crown for any tight areas which might be preventing the crown seating and can also be used to check the occlusion for premature contacts.
- *Contact areas*. These should be confirmed by using waxed floss. It is essential when checking the contact areas to secure the casting against the preparation with, for example, a firmly positioned ball-ended burnisher; holding the ball-ended burnisher against the casting is best carried out by a dental nurse. Check the contact areas by moving floss gingivally, not coronally as this will dislodge the restoration with potentially catastrophic consequences. The floss should meet with resistance when passing it through the contact area. Once checked, the floss can be



Figure 9.10 The Occlude-coated overlay is seated. The poor seating is evident as a large marginal gap. Where the contact point is tight, the Occlude spray has been worn away, indicating where the crown needs adjustment.





Figure 9.9 Occlude indicator spray has been sprayed onto the proximal surface of the gold cuspal overlay seen in Figures 9.5–9.8.



Figure 9.11 After adjusting the contact point on the overlay seen in Figures 9.5–9.10, the overlay seats down further.

removed from beneath the contact by pulling it through the interdental space in a buccal direction.

- *Interdental fit*. Again this is checked with waxed floss. If the margin has a positive ledge the floss will catch. Any overbuilt margins can be thinned first by using green stones and then abrasive discs and finally rubber points (Figure 9.12), all with decreasing abrasiveness. Final adjustment and polishing of accessible margins can be carried out after the restoration has been cemented onto the tooth but care should be exercised so as not to remove any gold bevel which would result in an unacceptably thick exposed cement lute.
- *Checking the occlusion*. This is carried out in a similar way to that detailed above when the restoration is checked on the cast (master die) prior to the patient's visit. Articulating paper can be used to mark up any premature contacts or high spots (Figure 9.13). Shimstock foil can be used to ensure that the restoration remains functional, in at least maximum intercuspation, if not other excursions depending on the occlusal scheme; the restoration should not be adjusted until it is not in occlusion (Figure 9.14).



Figure 9.12 Use of abrasive discs in order to achieve a satisfactory buccal margin. The grit marks are now removed with rubber cups/wheels before cementation.



Figure 9.13 Articulating paper being used in a Miller forceps to check the occlusion. Provisional restorations have been placed in the upper anterior region; the articulating paper is being used to check that there are posterior occlusal contacts as well as even occlusal contacts on the anterior provisional restorations.



Figure 9.14 Shimstock foil being used to check the occlusion on the provisional restorations seen in Figure 9.13. A tight occlusal contact exists between the premolar teeth which resist the withdrawal of the foil, hence the provisional restorations are not 'high'. When placed between the provisional restorations and the opposing teeth, resistance to withdrawal is also found, indicating that they are also in functional occlusion (inset).

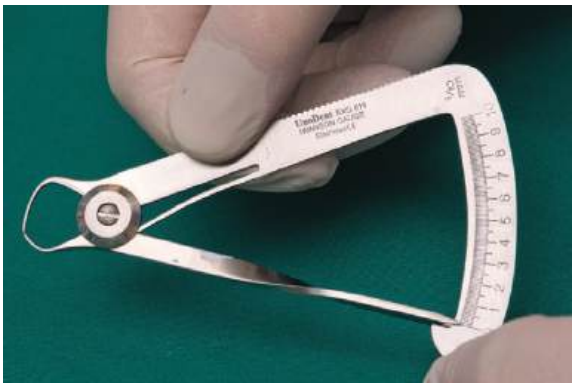


Figure 9.15 An Iwanson gauge. Ensure that when the tips of the gauge are in contact (left) the pointer reads zero (right).



Figure 9.16 An Iwanson gauge being used to measure the thickness of a gold crown. At the point in question the gold is 0.9 mm thick.

When adjusting the casting it is important that the gold does not become too thin (<0.3 mm) as there is a risk of perforation after years of function. An Iwanson gauge (UnoDent) is essential in determining the thickness of the crown material (Figures 9.15 and 9.16).

CEMENTATION OF THE RESTORATION

Before the gold restoration is prescribed the dentist must decide how it is to be cemented. This is because if adhesive cementation is required the correct alloy type needs to be specified. In any regard the cement used should be non-irritant and have sufficient working time to allow manipulation of the restoration into place. Cementation is helped if the material used is pseudoplastic, as such materials coat the fit surface well with no slumping, yet flow readily on seating the restoration home. Once set, the cement should seal the interface between the restoration and the tooth/core and demonstrate high compressive strength. In order to prevent microleakage at the restoration margins, it is also desirable that the cementing agent should have low solubility in the oral environment. Over the years a wide range of materials have been developed for this purpose. These include the following:

- *Zinc phosphate cements*. These cements are formed from the reaction of zinc oxide with phosphoric acid. They are generally presented as powder (ZnO) and liquid (H_3PO_4; 45–64%) and the key to their successful application is to observe the correct powder to liquid ratio upon mixing. Although successfully used over many years, they are non-adhesive and considered to be irritant to the pulp due to their initial high acidity and with time they are soluble in oral fluids. Whilst cheap and still widely available, zinc phosphate cements have been superseded by more modern adhesive materials and hence not discussed further in this text.
- *Polycarboxylate cements*. These arise from the reaction of zinc oxide powder with an aqueous solution of polyacrylic acid (30–40%). Some formulations have the acid freeze dried and added to the powder, which is then mixed with water. They are adhesive to tooth substance and form a weak bond to gold but not to porcelain. They are not considered to be irritant to the pulp. Clinical experience suggests that their higher solubility, compared to zinc phosphate, does not pose a significant clinical problem. The main disadvantages of these cements are that they have a short working time and crown placement needs to be quick. When mixed in the correct proportions, the cement initially appears too thick to flow and it is tempting to add more liquid to thin it down. This should be resisted as the cement is pseudoplastic and will flow well under pressure; in addition, adding more liquid will adversely affect the physical properties of the cement.
- *Glass polyalkenoate cements, e.g. AquaCem* (*Dentsply*). These cements are more commonly known as glass ionomers and the luting cement is derivative from the filling material of the same name. In common with these they are a reaction product of fluoroaluminosilicate glass and an aqueous solution of either acrylic acid or of a maleic/acrylic acid copolymer to which, in order to control the setting reaction, the manufacturer has incorporated tartaric acid. Some manufacturers, as with the polycarboxylate, have freeze dried the acids and added them to the powder, which is then mixed with water. It should be noted that in order to produce a film thickness amenable for luting, the particle size of the glass is less than in the filling material version. Such cements are adhesive to tooth substance and less soluble than the polycarboxylate cements. They are, however, susceptible to moisture contamination during the first few hours of placement. Any benefits of fluoride release and reduction in secondary caries are unlikely to play a significant clinical role (see Chapter 7).

- *Resin modified glass polyalkenoate cements and compomers, e.g. RelyX Luting Cement* (*3M ESPE*). These may simply be considered as a blend of glass ionomer and resin composite technology. They are less soluble than conventional glass polyalkenoate cements due to the incorporation of resin into these materials. Some concern has been expressed about the possibility of hygroscopic expansion in a moist environment. Whilst this is not a problem with metal-based restorations, its use with all-ceramic crowns should be avoided, perhaps with the exception of zirconia-based crowns.
- *Chemically adhesive luting resins, e.g. Panavia F* (*Kuraray Dental*) *and RelyX ARC* (*3M ESPE*). These are suitable for adhesive luting of Type 3 and 4 gold alloys. They derive from resin composites with the active constituent being either 4-META (4-methacryloxyethyl trimellitate anhydride) or MDP (10-methacryloyloxydecyldihydrogenphosphate). If oxidation of the gold restoration is enlisted to enhance its retention, the cemented, oxidized and blackened restoration must be polished by the dentist in situ. If metal-based crowns have been cemented with these cements, retrieval can be extremely difficult and principles of tooth preparation and conventions of retention and resistance form should be preferred.
- *Resin luting cements, e.g. RelyX Unicem* (*3M ESPE*). There is a distinction between these cements and the preceding group of cements in that this group does not bond chemically to metal. Some, like RelyX Unicem, are self-adhesive (etching and bonding) to tooth tissue and do not require a separate etching and bond application. These materials, like their restorative counterpart, also have good aesthetic properties if used in conjunction with indirect composite or ceramic restorations.

Luting the restoration

There is an abundance of in vitro product testing studies comparing the mechanical and biological properties of different luting cements. However, there are no known randomized controlled studies of sufficient duration to demonstrate that one cement results in a superior clinical outcome to another. Anecdote would suggest that a glass ionomer luting cement or resin-modified glass ionomer luting cement is appropriate for all metal-based restorations apart from those where the dentist considers the preparation lacks adequate retention and resistance properties. For these preparations an adhesive resin composite should be used.

Regardless of which luting material is chosen, the most important factor in carrying out this stage predictably is to follow the manufacturer's instructions. Other tips are:

- The dentist should repeatedly rehearse the path of insertion of the restoration.
- Complete moisture control with cotton wool rolls and aspiration – a saliva ejector or a Svendopter or Linguaflex suction and tongue retractor should be used (see Chapter 8). Rubber dam placement which allows a crown to be cemented unhindered is often difficult and unrealistic; a split dam may be the only option.
- The dental nurse mixes the correct ratio of materials according to manufacturer's instructions; no attempt should be made to alter this as it will adversely affect the properties of the material (Figure 9.17).
- With the restoration held by the dentist in the same manner as when establishing the correct path of insertion, the fit surface of the restoration is presented to the dental nurse.
- The nurse then carefully works/agitates the mixed cement over the entire fit surface (Figure 9.18) and the dentist seats the restoration.
- Before the cement is set, the occlusion is checked in intercuspal position to ensure complete seating.
- While the cement sets, the patient can gently close on a cotton wool roll placed between the restoration and the opposing tooth.
- When the cement is set partially, excess is removed (Figure 9.18).
- When fully set, the margins are examined meticulously. Every speck of excess cement is removed. Interdental excess cement is removed with dental floss.
- Accessible margins can be further finished with appropriate rubber points if necessary.
- If a glass ionomer luting cement is used, an unfilled resin can be applied to the margins to prevent any premature dissolution or drying out of the cement (Figure 9.18). This may be unnecessary with newer glass ionomer formulations.

Figures 9.19 and 9.20 show cemented gold crowns for patients illustrated in this and the preceding chapter.



Figure 9.17 A glass ionomer luting cement with water dispenser and scoop to ensure an accurate and reproducible ratio of powder (inset) to water.



Figure 9.18 From top, left to right. A gold full coverage bridge retainer is filled with a glass ionomer luting cement (the retainer only needs a complete lining), the bridge is completely seated, the excess cement allowed to reach the setting (gelation) stage and the excess gently removed with a probe. Excess interdental cement should always be removed by passing dental floss between the restoration and adjacent teeth. Finally, resin is applied to the margin to prevent premature dissolution of the cement.



Figure 9.19 Completed gold crown lower left first molar for patient seen in Figure 8.1B, Chapter 8. Note how the occlusal morphology has been refined to accommodate the group function and the gingival one-third has been polished to facilitate optimum oral hygiene.



Figure 9.20 Cemented full coverage gold crown for the patient seen in Figures 9.1–9.4.

SUPPORTIVE PHASE

Following the 'fit appointment', it is good practice to arrange a further visit to review the completed restoration. Not only can the restoration be reassessed after a period of function, but a dental carer can also encourage and reinforce oral health, lifestyle changes and aspects of home care that were discussed during the preparatory phase. Specifically, the patient should be asked whether or not they have been able to resume their normal home care, particularly interdental cleaning. Evidence of effective plaque removal and periodontal health should be confirmed by a clinical examination. At this appointment, it is occasionally necessary to refine the occlusion as the patient may have had local anaesthetic at the fit appointment and found it difficult to assess the occlusion accurately.

SUMMARY

Undergraduate students in many dental schools worldwide do not have the opportunity to make metal-based crowns for patients; however, knowledge of the materials and the laboratory techniques used is important. This knowledge will allow dentists to critically analyse these bespoke restorations, especially when castings do not appear to fit immediately. Whilst great care is taken over tooth preparation, impression taking and manufacture of metal crowns, little credence is given to cementation. This final stage in providing the patient with an indirect restoration is equally important. These important issues have been addressed throughout this chapter.

FURTHER READING

Burke, F.J., 2005. Trends in indirect dentistry: 3. Luting materials. Dent Update 32, 251–254, 257–258, 260.

Van Noort, R., 2007. Introduction to dental materials, third ed. Mosby, Edinburgh. Section 3.